

设计用户情境画像本体：以车辆销售领域为例 Designing a User Contextual Profile Ontology: A Focus on the Vehicle Sales Domain

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核心内容

Q：这篇论文解决的是什么问题？ A：一个人可以同时有好几个画像。论文举的例子是 Henri 要买两辆车，一辆公务用，一辆给带两个孩子的家庭用，于是 Henri 有两个画像，各有各的偏好。作者说以往研究要么只管用户画像，要么只管用户上下文，没人认真处理「每个画像各自带一份上下文」这件事。他们做的 UCPO 本体就是补这个缺口。

Q：一个人多个画像，在本体里怎么表达？ A：把上下文（Context）分成两个子类：UserContext 描述这个人整体的上下文，ProfileContext 描述某一个具体画像的上下文。这样 Henri 的公务画像和家庭画像各挂一份 ProfileContext，而「Henri 现在在什么位置、用什么设备」这类信息挂在 UserContext 上，不重复。上下文本身只有四个类：时间、地点、活动、设备。

Q：本体的需求是怎么定出来的？ A：用胜任力问题（competency questions）。作者列了一张表，8 条非形式化的问句，比如「用户的人口统计学信息是什么」「用户偏好的车型是什么」「用户的购车预算是多少」。这些问句的作用是识别本体里该有哪些概念、关系和属性，并且在最后的测试阶段，每一条问句要被转写成一条 SPARQL 查询（SPARQL 是查询 RDF 图数据的语言），跑出来的答案必须符合预期，本体才算过关。

Q：它用什么方法建本体？ A：SAMOD，一套把敏捷软件开发原则搬进本体工程的方法，分启动、设计与实现、测试与评估三个阶段，迭代推进。SAMOD 有一个关键手法叫概念小模型（concept modelelet）：先各自独立地建若干个描述领域某一侧面的小模型，故意不受上一轮已有模型的约束，最后再合并。作者据此把 UCPO 拆成两个小模型，第一个装基本静态的人口统计与社会信息，第二个装用户、画像、上下文三者之间的结构与关系。

Q：本体怎么验收？ A：三类测试，缺一不可。模型测试用 OOPS!、OntoQA、OntoMetrics 这类工具查整体一致性；数据测试是灌进真实实例三元组，看模型能不能把这些信息表达出来；查询测试是把胜任力问题转成 SPARQL，看能不能得到预期答案。论文明确写了一句：本体模型必须一直调，直到所有测试都通过。

Q：他们怎么衡量本体本身的质量？ A：用 OntoMetrics 跑出一组数字。基础指标是数量统计：38 个类、27 个对象属性、16 个数据属性、159 个实例、22 条子类公理。模式指标是形状统计：属性丰富度（每个

类平均几个属性) 0.42、继承丰富度 (每个类平均几个子类) 0.58、关系丰富度 (非继承关系占全部关系的比例) 0.55、公理与类之比 30.55。作者据此判断这个本体「在扁平 (横向) 与深层 (纵向) 之间取得了平衡」。

背景补充 / 点评

先交代几个术语, 读这篇论文会反复碰到。

SPARQL 是 W3C 定的、查询 RDF 图数据的标准语言, 语法上类似 SQL, 但查的是三元组图。论文里那段查询的意思是: 找到用户、找到该用户的画像、找到画像上的车辆偏好、再取出偏好里的品牌, 返回前 10 条。

OOPS! 全称 Ontology Pitfall Scanner, 是马德里理工做的在线本体缺陷扫描器, 能自动检出常见的建模错误 (比如定义了关系却没定义定义域和值域)。OntoMetrics 是一个把本体质量指标计算自动化的框架。这两个工具的存在说明一件事: 本体的一致性和形状可以机械检查, 不必全靠人看。

DL 表达力 ALH(D) 是描述逻辑 (Description Logic) 的一种记号, 用来标记这个本体用到了哪些逻辑构件。ALH(D) 指的是带角色层级 (H) 和数据类型 (D) 的属性语言 (AL), 属于表达力较弱的一档。表达力弱在这里是优点, 作者用它来支撑「这是一个轻量级知识图谱, 能塞进不同的应用架构里」这个判断。

我的判断: 这篇论文的车辆销售内容毫无迁移价值, 但它的三个方法论零件价值很高, 而且是四篇里跟我们现有做法对应得最紧的一篇。

第一个零件, 一个人多个画像。这一点比它看起来重要。多数画像系统的默认假设是「一个人一份画像」, 于是同一个人不同场景下互相矛盾的偏好会打起来——Henri 既要省油紧凑又要七座宽敞, 塞进一份画像里就是自相矛盾。论文的解法是承认矛盾不在数据里, 在于把两个场景压成了一份画像。UserContext 和 ProfileContext 分开这个设计, 把「哪些信息属于这个人」和「哪些信息属于这个人的某个身份」划开了。

第二个零件, 胜任力问题驱动本体, 并且最后要转成可执行的查询。这不是新鲜想法 (Ontology Development 101 那篇 2001 年的经典就在讲), 但这篇论文把它做完了闭环: 问句先写下来当需求, 本体照着建, 建完把问句转成 SPARQL 跑一遍验收。**问句不是文档, 是测试用例。**

第三个零件, 三类测试的分工。模型测试查结构自身有没有毛病, 数据测试查模型能不能装下真实数据, 查询测试查模型能不能答出该答的问题。三者不可替代: 结构再干净, 装不下真实数据也没用; 装得下数据, 答不出问题也没用。

留白与可疑之处也说清楚。这篇论文的评估非常薄: 那组 OntoMetrics 数字只能说明本体的形状, 说明不了它对不对。属性丰富度 0.42 是好还是坏? 论文说「高分表示设计质量高」, 但没给任何参照点, 也没跟别的本体比。案例研究那一节更像演示脚本——Louis 喜欢 Peugeot, 于是推荐 Peugeot 206/207/208——没有任何指标, 没有对照组, 读不出这个本体到底带来了什么。所以这篇的可信部分是方法, 不是效果。

对我自己的启示, 有四条, 其中两条是确认, 两条是新东西。

先说两条确认的。第一，它的胜任力问题加 SPARQL 验收，和本仓库的 CQ 注册表加评测是同一个结构：问句先登记，指标照着建，最后用评测跑通才转 accepted。这篇论文让我确认那套做法不是我们自创的土办法，它在本体工程里是标准动作。第二，它的三类测试和我们现在的验证顺序也对得上：加载期引用完整性校验对应模型测试，dev 统一数仓上跑真实数据对应数据测试，十问验收对应查询测试。三者缺一不可这个判断，我们是踩过坑才知道的（591 条单测全绿也证明不了那十句还答得对），论文是直接写出来的。

再说两条新东西。

第三条，一个学员应该允许有多份画像。这一条现在最该拿去用。Learning Lab 要做「切己迁移」，就得知道学员当下关注什么；而同一个人「作为团队管理者」和「作为销售个人贡献者」关注的东西不同，压成一份画像会互相污染。按 UCPO 的分法，稳定的个人信息（岗位、年龄、教育）挂在人身上，关注点与偏好挂在具体画像上，每个画像各自带上下文。这也顺带给「愿景到底该怎么存」提供了一条路：愿景挂在画像上，不挂在人上，因为一个人在不同身份下的愿景可以不同。

第四条，本体的形状可以量化，用来防畸形。属性丰富度、继承丰富度、关系丰富度这三个数字，各自的绝对值没什么意义（论文也没给参照点），但它们的**变化趋势**有意义：如果画像本体迭代三个月后，继承丰富度一路上升而关系丰富度一路下降，说明我们在不停地加子类分层、却没在加关系，本体正在变成一棵越来越深的分类树。分类树是标签体系，不是本体。这个信号靠人看类图看不出来，靠三个数字能看出来。

文中金句

"The ontology model must be adjusted until all tests are successful." 本体模型必须一直调整，直到所有测试都通过。

"Therefore, the user Henri can have two distinct profiles, each catering to different needs and preferences." 因此，用户 Henri 可以拥有两个截然不同的画像，各自服务于不同的需要与偏好。

"Competency questions serve as a guide for ontology development and help to ensure that the resulting ontology accurately represents the user context and profile information required for personalization." 胜任力问题为本体开发提供指引，并帮助确保最终得到的本体准确表示了个性化所需的用户上下文与画像信息。

"These modelets are used to provide an initial conceptualization without being constrained by the current model available after the previous iteration of the process." 使用这些概念小模型的目的是给出一个初始的概念化，而不受上一轮迭代后已有模型的约束。

摘要

In the digital age, it is crucial to understand and tailor experiences for users interacting with systems and applications. This requires the creation of user contextual profiles that combine user profiles with contextual information. However, there is a lack of research on the integration of contextual information with different user profiles. This study aims to address this gap by designing a user contextual profile ontology that considers both user profiles and contextual information on each profile.

在数字时代，理解与系统和应用交互的用户、并为其定制体验，是至关重要的。这需要创建「用户情境画像」，即把用户画像与上下文信息结合起来。然而，关于「把上下文信息与用户的不同画像整合起来」这件事，研究是缺失的。本研究旨在填补这一缺口，设计一个用户情境画像本体，同时考虑用户画像本身、以及每一个画像各自的上下文信息。

Specifically, we present a design and development of the user contextual profile ontology with a focus on the vehicle sales domain. Our designed ontology serves as a structural foundation for standardizing the representation of user profiles and contextual information, enhancing the system's ability to capture user preferences and contextual information of the user accurately. Moreover, we illustrate a case study using the User Contextual Profile Ontology in generating personalized recommendations for vehicle sales domain.

具体而言，我们给出一个聚焦车辆销售领域的用户情境画像本体的设计与开发过程。我们设计的本体为「用户画像与上下文信息的表示标准化」提供了结构性基础，提升了系统准确捕捉用户偏好与上下文信息的能力。此外，我们用一个案例研究说明如何用这个用户情境画像本体，为车辆销售领域生成个性化推荐。

一、引言

Understanding and customizing experiences for users interacting with various systems and applications is essential in today's digital landscape. To accomplish this, individual user profiles are integrated with their corresponding contextual information, resulting in the creation of a user contextual profile. A user profile is a compilation of personal information and preferences about a particular user, including demographic data, interests, past behavior, and other attributes that define the user's identity within a system or application.

在当今的数字环境中，理解并为与各种系统和应用交互的用户定制体验，是必不可少的。为达成这一点，需要把单个用户画像与其对应的上下文信息整合起来，从而形成用户情境画像。所谓用户画像，是关于某个特定用户的个人信息与偏好的汇编，包括人口统计学数据、兴趣、过往行为，以及在某个系统或应用中定义该用户身份的其他属性。

For example, the user Henri wants to buy a vehicle for professional use and another for his family with two children. Therefore, the user Henri can have two distinct profiles, each catering to different needs and preferences. While contextual information for each profile refers to the situational factors surrounding a user's interaction with a system or application. This may include location, time, device being used, and even the user's current activity or emotional state.

举例来说，用户 Henri 想买一辆车用于公务，再买一辆给有两个孩子的家庭用。因此，用户 Henri 可以拥有两个截然不同的画像，各自服务于不同的需要与偏好。而每个画像的上下文信息，指的是围绕该用户与系统或应用交互的那些情境因素，可能包括地点、时间、所用设备，甚至用户当前的活动或情绪状态。

Consequently, a user contextual profile combines the user profile with the contextual information specific to that profile, creating a more comprehensive understanding of the user's needs and preferences in various situations. For instance, when searching for a professional vehicle, the user contextual profile may prioritize factors such as fuel efficiency, compact size, and reliability, while for a family vehicle, it may focus on safety features, ample seating, and cargo space.

因此，用户情境画像把用户画像与该画像特有的上下文信息结合起来，从而对用户在各种情形下的需要与偏好形成更完整的理解。举例来说，在寻找公务用车时，用户情境画像可能会优先考虑省油、尺寸紧凑、可靠性这些因素；而对家庭用车，则可能聚焦安全配置、充足座位和行李空间。

In order to effectively model user contextual profiles, utilizing ontology-based approaches can be advantageous. Ontologies are formal, structured representations of knowledge within a specific domain that enable the organization and sharing of data in a machine-readable format. Once the ontology is designed and implemented, it can be integrated into applications and systems, allowing them to reason over the user contextual profiles and infer new knowledge based on the available data. This can improve the accuracy and relevance of recommendations, as well as facilitate more efficient data integration and interoperability among different applications and services.

为了有效地给用户情境画像建模，采用基于本体的方法是有利的。本体是特定领域内知识的形式化、结构化表示，使数据能以机器可读的格式被组织和共享。本体一旦设计并实现，就能集成进应用与系统，让它们能在用户情境画像上做推理，并基于已有数据推出新知识。这既能提升推荐的准确性与相关性，也能促进不同应用与服务之间更高效的数据集成与互操作。

A considerable number of studies in the field of user modeling have primarily focused on either user profiles or user contexts. The aspects of contextual information related to each profile of the user have not been thoroughly examined. This underexplored area offers an opportunity for further research, focusing on the integration of contextual information with different profiles of the user to develop a deeper understanding of users' needs and preferences in various situations.

用户建模领域相当多的研究，主要要么聚焦用户画像，要么聚焦用户上下文。与用户的每一个画像相关的上下文信息这一侧面，并没有被彻底考察过。这个探索不足的领域为进一步研究提供了机会：聚焦于把上下文信息与用户的不同画像整合起来，以对用户在各种情形下的需要与偏好形成更深的理解。

二、相关工作

2.1 本体化的用户建模

Ontologies have proven to be highly effective in modeling user profiles and contexts. They provide a thorough depiction of a particular domain of interest, simplifying browsing and query refinement. Specifically, ontologies have been observed to outperform other methods in user modeling when compared to other methods utilized. Ontological user modeling refers to the use of ontologies for representing and managing user profiles, preferences, and contextual information in a structured and machine-readable format.

本体已被证明在给用户画像和上下文建模方面非常有效。它们对某个特定的关注领域给出全面的刻画，简化了浏览与查询精化。具体而言，与其他被采用的方法相比，本体在用户建模上的表现被观察到优于它们。本体化的用户建模，指的是用本体以结构化、机器可读的格式来表示和管理用户画像、偏好与上下文信息。

User-related information such as profiles, preferences, and context are represented using concepts, relationships, and attributes within a formal, hierarchical structure. This structured representation enables more effective processing, reasoning, and inferencing by computer systems. Ontological user modeling has previously been investigated across numerous research areas, demonstrating its adaptability and potential to enhance user experiences in diverse domains such as e-learning, e-commerce, recommender systems, personalized web services, and context-aware applications.

与用户相关的信息，比如画像、偏好、上下文，都在一个形式化的层级结构中用概念、关系和属性来表示。这种结构化表示使计算机系统能更有效地处理、推理与推断。本体化的用户建模此前已在众多研究领域被考察过，在在线学习、电子商务、推荐系统、个性化 Web 服务、上下文感知应用等多个领域展现出适应性与提升用户体验的潜力。

Golemati et al. designed an ontology that incorporates concepts and properties essential for modeling user profiles. Their model predominantly focuses on static user characteristics and provides a foundation for the development of a more general, comprehensive, and extensible user model. Adewoyin et al. proposed a versatile user modeling architecture for smart environments. They represented the characteristics of users using the behavior concept. Sutterer et al. put forth a user profile ontology designed specifically to depict situation-dependent sub-profiles. Skillen et al. developed a User Profile Ontology for personalizing context-aware applications in mobile environments. Their focus lies on user behavior and characterizing the needs of users for context-aware applications.

Golemati 等人设计了一个本体，纳入了给用户画像建模所必需的概念与属性。他们的模型主要聚焦静态的用户特征，为开发更通用、更完整、更可扩展的用户模型提供了基础。Adewoyin 等人提出了一套面向智能环境的通用用户建模架构，他们用「行为」这一概念来表示用户特征。Sutterer 等人提出了一个专门用来刻画「依情形而定的子画像」的用户画像本体。Skillen 等人开发了一个用户画像本体，用于在移动环境中个性化上下文感知应用，他们的重点在用户行为，以及刻画用户对上下文感知应用的需要。

The works mentioned highlight numerous advantages of ontological user modeling, which is capable of capturing both static and dynamic user characteristics related to permanent, temporary information, and user evolution. Nonetheless, the contextual information can be dependent on the profile and its primary objectives. Therefore, it is necessary to distinguish contextual information for the user from that for the profile clearly.

上述工作凸显了本体化用户建模的诸多优点：它既能捕捉静态的、也能捕捉动态的用户特征，涵盖长期信息、临时信息以及用户的演变。尽管如此，上下文信息可能依赖于具体画像及其主要目标。因此，有必要把「属于用户的上下文信息」和「属于画像的上下文信息」清楚地区分开。

2.2 本体开发方法论

Ontology development methodologies provide a structured and systematic approach to the creation, maintenance, and evolution of ontologies. These methodologies are specifically designed to ensure the quality, coherence, and practicality of the resulting ontology, while also promoting its consistency and usability.

本体开发方法论为本体的创建、维护与演化提供结构化、系统化的路径。这些方法论专门用于确保最终本体的质量、融贯性与实用性，同时提升其一致性与可用性。

Various ontology development methodologies have been proposed in the literature, each featuring its own unique set of guidelines and principles. Some widely adopted methodologies include: Methontology, Ontology Development 101 (OD101), NeON Methodology, Unified Process for Ontology Building (UPON), Simplified Agile Methodology for Ontology Development (SAMOD), Modular Ontology Modeling (MoMo), and others.

文献中提出了多种本体开发方法论，各有自己独特的一套指南与原则。一些被广泛采用的方法论包括：Methontology；本体开发 101（OD101）；NeON 方法论；本体构建统一过程（UPON）；本体开发简化敏捷方法论（SAMOD）；模块化本体建模（MoMo）等等。

Firstly, Methontology is one of the earliest and most widely used methodologies. It covers the entire ontology development process, including specification, conceptualization, formalization, integration, implementation, and maintenance, while emphasizing the importance of documentation throughout the development process. OD101 then is a methodology designed to guide ontology engineering processes. Focusing on iterative development, OD101 provides step-by-step guidelines and best practices for constructing ontologies.

首先，Methontology 是最早且使用最广泛的方法论之一。它覆盖本体开发的全过程，包括规格说明、概念化、形式化、集成、实现与维护，同时强调开发全过程中文档的重要性。其次，OD101 是一套用于指导本体工程流程的方法论。它聚焦迭代式开发，为构建本体提供逐步的指南与最佳实践。

The NeON Methodology next is designed to create ontology networks, rather than singular ontologies. This approach emphasizes the reuse and re-engineering of existing ontologies, fostering collaboration between ontology engineers and domain experts. UPON is a methodology that incorporates concepts from the Unified Modeling Language (UML) and the Rational Unified Process (RUP). This approach adheres to an iterative and incremental process, encompassing four development phases: inception, elaboration, construction, and transition.

再者，NeON 方法论的设计目标是创建本体网络，而不是单个本体。这一路径强调对已有本体的复用与再工程，促进本体工程师与领域专家之间的协作。UPON 则是一套吸收了统一建模语言（UML）与统一软件开发过程（RUP）概念的方法论。它遵循迭代增量的流程，包含四个开发阶段：起始、细化、构造、移交。

SAMOD is a methodology that integrates agile software development principles into ontology engineering. It provides a flexible, iterative, and adaptable framework, fostering collaboration among ontology engineers, domain experts, and end-users. Employing iterative development cycles, SAMOD encompasses planning, design, implementation, and evaluation stages, facilitating ongoing feedback and refinements throughout the development process. Finally, MoMo is a methodology that creates and manages modular ontologies, allowing for independent development, maintenance, and reuse of smaller, more manageable modules.

SAMOD 是一套把敏捷软件开发原则融入本体工程的方法论。它提供灵活、迭代、可适应的框架，促进本体工程师、领域专家与最终用户之间的协作。SAMOD 采用迭代开发周期，涵盖规划、设计、实现、评估四个阶段，使开发全过程中持续的反馈与精化成为可能。最后，MoMo 是一套创建并管理模块化本体的方法论，允许对更小、更易管理的模块做独立的开发、维护与复用。

The selection of an appropriate ontology development methodology depends on several factors, such as the complexity of the ontology, the domain of application, the availability of resources, and the expertise of the ontology engineers. In this work, we have opted for SAMOD to build a user contextual profile ontology due to its flexibility, adaptability, and emphasis on collaboration among ontology engineers, domain experts, and end-users.

选择哪一套本体开发方法论，取决于若干因素，例如本体的复杂度、应用领域、资源可得性，以及本体工程师的专长。在这项工作中，我们选择用 SAMOD 来构建用户情境画像本体，理由是它灵活、可适应，并且强调本体工程师、领域专家与最终用户之间的协作。

三、开发用户情境画像本体的方法

3.1 启动阶段

The main objective of the UCPO is to establish a standardized representation of user information and contextual information for each user profile. Defining a standardized format for user information, context, and profile can improve the interoperability of the UCPO and enable it to adapt to the changing needs and preferences of the users. The scope of the UCPO is broad, covering a wide range of user characteristics and environmental factors that can influence user behavior and preferences. This includes, but is not limited to, demographics, psychographics, device and network context, physical context, social context, and temporal context.

UCPO 的主要目标是为「用户信息」以及「每个用户画像各自的上下文信息」建立一种标准化表示。为用户信息、上下文和画像定义标准化格式，能提升 UCPO 的互操作性，并使其能适应用户不断变化的需要与偏好。UCPO 的范围很广，覆盖能影响用户行为与偏好的各类用户特征与环境因素，包括但不限于：人口统计学特征、心理特征、设备与网络上下文、物理上下文、社会上下文、时间上下文。

To develop the UCPO, we consult a variety of information sources and references such as academic articles, conference proceedings, textbooks, and online resources related to ontology engineering, user modeling, and personalization. In selecting the information sources, it is important to consider their relevance to the scope and goals of the UCPO, as well as the quality and credibility of the sources. We manually reviewed and analyzed the relevant literature to identify the key concepts and terms related to user context and personalization. Additionally, consulting domain experts, ontology engineers, and end-users can provide valuable insights.

为开发 UCPO，我们查阅了各种信息来源与参考资料，例如与本体工程、用户建模、个性化相关的学术论文、会议论文集、教科书和在线资源。在挑选信息来源时，既要考虑它们与 UCPO 范围和目标的相关性，也要考虑来源本身的质量与可信度。我们人工回顾并分析了相关文献，以识别与用户上下文和个性化相关的关键概念与术语。此外，咨询领域专家、本体工程师和最终用户也能提供有价值的洞见。

The ontology requirements can be formulated by using informal competency questions. The set of competency questions helps to identify and define the key concepts, relationships, and attributes that should be included in the ontology. Competency questions serve as a guide for ontology development and help to ensure that the resulting ontology accurately represents the user context and profile information required for personalization.

本体的需求可以用非形式化的胜任力问题来表达。这一组胜任力问题帮助识别并定义本体中应当包含的关键概念、关系与属性。胜任力问题为本体开发提供指引，并帮助确保最终得到的本体准确表示了个性化所需的用户上下文与画像信息。

（原文表 1 给出车辆销售领域的示例胜任力问题：CQ1 用户的人口统计学信息是什么？CQ2 用户偏好的车型是什么？CQ3 用户的购车预算是多少？CQ4 用户偏爱哪些具体车型？CQ5 用户的驾驶环境是什么？CQ6 用户偏好的汽车品牌是什么？CQ7 某一具体车型的主要使用场景有哪些？CQ8 用户偏好的变速箱类型是什么？）

Defining the goal, scope, information sources, and competency questions helps to establish a shared understanding of the primary requirements of the UCPO between domain experts and

ontology engineers.

定义目标、范围、信息来源和胜任力问题，有助于在领域专家与本体工程师之间，就 UCPO 的主要需求建立共同理解。

3.2 设计与实现阶段

Considered as the core of the ontology engineering process, the design and implementation phase consists of several iterations of ontology design, implementation, and testing until a comprehensive and adaptable ontology is achieved. The design phase begins by identifying the key concepts and relationships that should be included in the ontology based on the competency questions formulated in the previous phase. SAMOD prescribes an iterative process that aims to build the final model through a series of small steps. One key component of this process is the use of concept modelets, which are standalone models that describe particular aspects of the domain under consideration. These modelets are used to provide an initial conceptualization without being constrained by the current model available after the previous iteration of the process.

设计与实现阶段被视为本体工程流程的核心，它由若干轮本体设计、实现与测试的迭代组成，直到得到一个完整且可适应的本体。设计阶段的起点是：基于上一阶段所拟定的胜任力问题，识别本体中应当包含的关键概念与关系。SAMOD 规定了一个迭代流程，目标是通过一系列小步骤构建出最终模型。这个流程的一个关键组成部分是使用「概念小模型」，即描述所考察领域某个特定侧面的独立模型。使用这些小模型的目的是给出一个初始的概念化，而不受上一轮迭代后已有模型的约束。

We have organized the development of the UCPO into two separate modelets. The first modelet focuses on representing mostly static and permanent user characteristics. The second modelet presents both static and dynamic information about the user profile and contextual factors that affect the user and their profile. The first modelet is responsible for organizing user demographic and social information, such as Gender, Age, Address, Occupation, Income, Language, NumberOfChildren, MaritalStatus, Education, and more.

我们把 UCPO 的开发组织成两个独立的小模型。第一个小模型侧重表示主要是静态、长期的用户特征。第二个小模型同时呈现关于用户画像的静态与动态信息，以及影响用户及其画像的上下文因素。第一个小模型负责组织用户的人口统计学与社会信息，例如性别、年龄、地址、职业、收入、语言、子女数量、婚姻状况、教育程度等等。

Occupation refers to a person's job or profession, which can provide insights into their daily routine and transportation needs. Income refers to a person's financial status, which can affect their purchasing power and preferences. NumberOfChildren refers to the number of children a person has, which can impact their vehicle preferences and needs. In our research, we utilized the work of Bermudez et al. for the basic classes of user information. Moreover, we have

created a class named `PersonalProfile`, which acts as a superclass for the classes related to the demographic and social information.

职业指一个人的工作或专业，能透露其日常作息与交通需求。收入指一个人的财务状况，会影响其购买力与偏好。子女数量指一个人有几个孩子，会影响其对车辆的偏好与需要。在我们的研究中，用户信息的基础类采用了 Bermudez 等人的成果。此外，我们创建了一个名为 `PersonalProfile` 的类，作为与人口统计学和社会信息相关的那些类的父类。

The second model focuses on the representation of the structure and relationships between a user, their user profile, and their user context. User context refers to the circumstances, environment, and situation in which a user interacts with a system or application. We investigated four particular contextual information about the user, including Time, Location, Activity, and Device classes. In our design, we created two subclasses of Context class: the `UserContext` class that describes general contextual information about the user and the `ProfileContext` class, which describes contextual information for each user profile.

第二个小模型侧重表示用户、用户画像、用户上下文三者之间的结构与关系。用户上下文指的是用户与系统或应用交互时所处的境况、环境与情形。我们考察了四类具体的用户上下文信息，即时间、地点、活动、设备四个类。在我们的设计中，我们为 Context（上下文）类创建了两个子类：`UserContext` 类描述关于用户的一般性上下文信息；`ProfileContext` 类描述每一个用户画像各自的上下文信息。

Furthermore, we have provided the ability for each user to declare their preferences and create various types of profiles that align with their intended purpose while interacting with the system or application. To structure this type of information, we have created two classes: `Preference` and `UserProfile`. In order to address the needs and preferences of users in the vehicle sales domain, we have created a set of preferences that relate to the user's desired vehicle. This set of preferences includes several classes such as `VehicleType`, `RouteType`, `Mileage`, `Color`, `NumberOfPlaces`, `State`, `Budget`, and `Brand`.

此外，我们让每个用户都能声明自己的偏好，并创建与其在与系统或应用交互时的意图目的相符的多种画像。为组织这类信息，我们创建了两个类：`Preference`（偏好）与 `UserProfile`（用户画像）。为应对车辆销售领域中用户的需要与偏好，我们创建了一组与用户想要的车辆相关的偏好，包含车型、路线类型、里程、颜色、座位数、车况、预算、品牌等若干类。

It is important to note that depending on the specific domain and application, additional preferences may be necessary to fully capture the user's needs and preferences. Therefore, sub-classes of `Preference` class can be created and customized based on the specific requirements of the vehicle sales domain and the user base.

需要注意的是，依据具体领域与应用的不同，可能需要额外的偏好才能完整捕捉用户的需要与偏好。因此，可以基于车辆销售领域和用户群体的具体要求，创建并定制 `Preference` 类的子类。

The implementation of the UCPO involves the process of actually creating and deploying the ontology in a system or application. This requires converting the conceptual model of the ontology into a computer-readable format using ontology languages such as OWL or RDF. We have implemented the UCPO using the Web Ontology Language (OWL) with the support of the Protégé-OWL editor.

UCPO 的实现，涉及在系统或应用中真正创建并部署本体这一过程。这要求用 OWL 或 RDF 这类本体语言，把本体的概念模型转换成计算机可读的格式。我们借助 Protégé-OWL 编辑器，用 Web 本体语言（OWL）实现了 UCPO。

3.3 测试与评估阶段

In the final phase of the UCPO development, we focus on testing and evaluating the produced ontology. The tests can be categorized into a model test, a data test, and query tests. To perform the model test, we can employ evaluation metrics such as OOPS!, OntoQA, and OntoMetrics to verify the overall consistency of the developed ontology. The data test consists of checking the validity of the model after populating it with instance triplets. Finally, for query tests, informal competency questions must be transformed into SPARQL queries to ensure that the expected answers are obtained. The ontology model must be adjusted until all tests are successful.

在 UCPO 开发的最后阶段，我们聚焦于对产出的本体做测试与评估。测试可分为模型测试、数据测试和查询测试三类。做模型测试时，我们可以使用 OOPS!、OntoQA、OntoMetrics 这类评估指标工具，来验证所开发本体的整体一致性。数据测试的内容是：在给模型灌入实例三元组之后，检查模型的有效性。最后，查询测试要求把非形式化的胜任力问题转换成 SPARQL 查询，以确保能得到预期的答案。本体模型必须一直调整，直到所有测试都通过。

In our work, we evaluated the quality of our ontology by examining its structure. Specifically, we observed several metrics using the OntoMetrics framework. The base metrics consist of simple metrics that count the number of axioms, classes, object properties, data properties, and individual instances. These metrics provide statistics about the ontology with regard to the quantity of elements present within it. Based on these statistics, we can conclude that our ontology is a lightweight knowledge graph that can be adopted in different application architectures. Moreover, the Description Logic (DL) expressivity of the ontology is ALH(D).

在我们的工作中，我们通过考察本体的结构来评估其质量。具体来说，我们用 OntoMetrics 框架观测了若干指标。基础指标由一些简单的计数指标构成，统计公理数、类数、对象属性数、数据属性数和个体实例数。这些指标给出关于本体内部元素数量的统计。基于这些统计，我们可以得出结论：我们的本体是一个轻量级知识图谱，能够被采纳进不同的应用架构。此外，该本体的描述逻辑（DL）表达力为 ALH(D)。

（原文表 2 的数值：类数 38；对象属性数 27；数据属性数 16；属性总数 43；个体实例数 159；子类公理数 22；对象属性定义域公理数 28；对象属性值域公理数 33；DL 表达力 ALH(D)。模式指标：属性丰富

度 0.421053; 继承丰富度 0.578947; 关系丰富度 0.55102; 公理/类之比 30.552632; 类/关系之比 0.77551。)

In schema metrics, the evaluation of the ontology focuses on the measure of attribute, inheritance, and relationship richness. Particularly, the attribute richness (AR) is considered as the average number of attributes per class, computed as the number of attributes for all classes divided by the number of classes. The high score of the attribute richness indicates the high quality of ontology design and the amount of information belong to instances. The inheritance richness (IR) is defined as the average number of sub-classes per class. The inheritance richness allows describing the distribution of information on different levels of the ontology. The relationship richness (RR) is defined as the percentage of the relationships between classes, computed as the number of non-inheritance relationships divided by the sum of inheritance and non-inheritance relationships.

在模式指标方面，对本体的评估聚焦于属性丰富度、继承丰富度、关系丰富度这三项度量。其中，属性丰富度（AR）被视为每个类平均拥有的属性数，计算方式是所有类的属性总数除以类的总数。属性丰富度得分高，表明本体设计质量高，且实例所承载的信息量大。继承丰富度（IR）定义为每个类平均拥有的子类数，它能刻画信息在本体不同层级上的分布。关系丰富度（RR）定义为类之间关系的占比，计算方式是非继承关系数除以继承关系数与非继承关系数之和。

Based on the explanation of metrics, these measured results indicate that our proposed ontology is balanced between a horizontal (or shallow) ontology and vertical (or deep) ontology.

基于上述指标的解释，这些测量结果表明我们提出的本体在「横向（扁平）本体」与「纵向（深层）本体」之间取得了平衡。

To conduct a data test for the UCPO, we need to check the validity of the model after populating it with instance triplets. For example, let's assume we have created instances for Henri's two profiles: his professional profile and his family profile. We can populate these instances with data such as his occupation and work location for his professional profile, and his number of children and preferred family activities for his family profile. This ensures that the UCPO can capture and express this information in a structured and consistent manner.

要对 UCPO 做数据测试，我们需要在给模型灌入实例三元组之后检查其有效性。举例来说，假设我们为 Henri 的两个画像创建了实例：公务画像和家庭画像。我们可以给公务画像灌入他的职业和工作地点这类数据，给家庭画像灌入他的子女数量和偏好的家庭活动。这样能确保 UCPO 能以结构化、一致的方式捕捉并表达这些信息。

In order to conduct a query test on the UCPO, we can convert the set of informal competency questions into SPARQL queries to ensure that the expected answers are obtained. For instance, if the competency question is "What is the user's preferred vehicle brand?", a suitable SPARQL query can be formulated to search for the first 10 users and their favorite brands, by traversing from user to user profile, to vehicle preference, to favorite brand.

要对 UCPO 做查询测试，我们可以把那组非形式化的胜任力问题转换成 SPARQL 查询，以确保能得到预期答案。举例来说，如果胜任力问题是「用户偏好的汽车品牌是什么」，那么可以写出一条相应的 SPARQL 查询：沿着「用户 → 用户画像 → 车辆偏好 → 偏爱品牌」这条路径遍历，取出前 10 个用户及其偏爱的品牌。

四、案例研究：个性化推荐

To demonstrate the application of the UCPO, we conducted a case study on personalized recommendations in the vehicle sales domain. Using the ontology, we were able to capture the user's contextual information, preferences, and profiles to provide tailored recommendations for vehicle purchases. Specifically, scenarios were considered where a user created a profile specifying their preferred vehicle type, route type, mileage, color, number of seats, location, brand, and budget. Additionally, the user's profile context included their activity, time of day, and device.

为演示 UCPO 的应用，我们在车辆销售领域做了一个关于个性化推荐的案例研究。借助该本体，我们能够捕捉用户的上下文信息、偏好和画像，从而为购车提供定制化推荐。具体而言，我们考虑了这样的场景：用户创建一个画像，在其中指定偏好的车型、路线类型、里程、颜色、座位数、地点、品牌和预算；此外，该用户的画像上下文包含其活动、时段和设备。

Louis and Pierre are two potential buyers of used vehicles who have registered on a vehicle sales application and set up their profiles. Louis has specified his preferences while expressing a particular interest in sedan models. On the other hand, Pierre has emphasized the importance of safety features as a top priority due to frequent travel with his family. By leveraging Louis's contextual information and preferences, including the Peugeot models he has previously liked, the application recommends models such as Peugeot 206, Peugeot 207, or Peugeot 208. For Pierre, who previously owned a Toyota and expressed interest in hybrid vehicles, the recommendation engine suggests a Toyota RAV4 Hybrid, a Honda CR-V Hybrid, and a Lexus UX Hybrid, taking into account factors such as safety ratings, fuel efficiency, and cost.

Louis 和 Pierre 是两位二手车的潜在买家，他们在一个车辆销售应用上注册并建立了自己的画像。Louis 指定了他的各项偏好，并表达出对轿车车型的特别兴趣。而 Pierre 由于经常与家人一同出行，强调把安全配置作为首要考量。应用利用 Louis 的上下文信息与偏好（包括他此前点赞过的标致车型），推荐了标致 206、标致 207、标致 208 这些车型。对于 Pierre，他此前拥有一辆丰田并对混动车表达过兴趣，推荐引擎给出了丰田 RAV4 混动、本田 CR-V 混动、雷克萨斯 UX 混动，并把安全评级、燃油经济性和价格等因素考虑在内。

Consider the example of Henri, who has two distinct profiles: one as a professional profile and the other as a family profile. By leveraging the UCPO, which captures user profiles and contextual information of each profile, relevant recommendations can be provided that cater to each corresponding profile. Specifically, for Henri's professional profile who has shown interest in Renault models, vehicles such as Renault Megane and Renault Talisman, which have spacious

trunks, advanced safety features, and good fuel efficiency, could be recommended. For Henri's family profile, who has indicated interest in SUV vehicle types, vehicles such as Koleos SUV and Renault Scenic could be recommended.

再看 Henri 的例子，他有两个截然不同的画像：一个是公务画像，一个是家庭画像。借助能同时捕捉用户画像与每个画像各自上下文信息的 UCPO，就能给出适配各自画像的相关推荐。具体来说，Henri 的公务画像表现出对雷诺车型的兴趣，那么后备箱宽敞、安全配置先进、油耗表现好的雷诺梅甘娜、雷诺 Talisman 就可以被推荐；而 Henri 的家庭画像表明其对 SUV 车型有兴趣，那么科雷傲 SUV 和雷诺风景这类车型就可以被推荐。

五、结论与展望

In this paper, we have presented the design and development process of the User Contextual Profile Ontology with a focus on the vehicle sales domain. This work established the bridge for the gap in the representation of user profile and user context by designing a user contextual profile ontology that considers both user profiles and contextual information on each profile. It is worth emphasizing that the development of the UCPO follows the main phases of ontology engineering, ensuring its consistency, validity, and effectiveness in achieving its intended purpose.

在本文中，我们给出了聚焦车辆销售领域的用户情境画像本体的设计与开发过程。这项工作通过设计一个同时考虑用户画像本身与每个画像各自上下文信息的本体，为「用户画像与用户上下文的表示」这一缺口搭起了桥梁。值得强调的是，UCPO 的开发遵循了本体工程的主要阶段，从而确保了它的一致性、有效性，以及达成其预期目的的效力。

In future work, we plan to extend the ontology to include additional information sources, such as user reviews and social media feeds, to enhance the accuracy of personalized recommendations. Finally, we aim to collaborate with industry partners to integrate the UCPO into commercial applications and evaluate its effectiveness in real-world settings.

在未来工作中，我们计划扩展这个本体，纳入更多信息来源，例如用户评论和社交媒体信息流，以提升个性化推荐的准确性。最后，我们希望与产业伙伴合作，把 UCPO 集成进商业应用，并在真实场景中评估其有效性。

参考链接

- [论文原文 arXiv:2308.06018](#) — 本文原始出处
- [OOPS! 本体缺陷扫描器](#) — 论文模型测试环节使用的在线工具，自动检出常见建模错误
- [User Profile Ontology \(CityPulse 项目\)](#) — 论文复用的用户信息基础类的出处，对应文中 up: 前缀
- [SAMOD 方法论原文](#) — Peroni 提出的本体开发简化敏捷方法论，本文所用流程的出处